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Problem Solving Protocol

Auditing cross-cohort feature-selection benchmarks: cohort composition and tissue-context-dependent confounding across kidney, liver and colon transcriptomes

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Abstract

We present a reusable audit protocol that quantifies how far cross-cohort feature-selection benchmarks depend on cohort composition and control definition, and weighs confounder correction against its costs. Across 26 subsets of five diabetic kidney disease cohorts, the reproducibility difference between a stability selector and a co-expression hub criterion ranged from -0.106 to $+0.170$ over three-cohort subsets, while the stability selector's mean external AUROC advantage stayed positive: one configuration does not settle the verdict. Instability tracked how far cohorts agreed about effects; subsampling colon cohorts to kidney sizes gave only rare, small reversals. The most reproducible signal was an immediate-early gene module tracking biopsy versus surgical procurement in 8 of 9 kidney datasets rather than disease. A biopsy control retained 14–56% of the naive signal across seven diagnoses; residualising on a handling score cut procurement-driven genes from 27% to 5% with no AUROC loss detected, on an interval too wide to establish preserved performance, whereas batch correction by cohort was inert; only 14 of 30 candidates survived correction. In liver and colon, under predictions fixed in advance, the module was lower in biopsy cases than surgical controls in 9 of 10 such cohorts and most prominent among reproducible genes in kidney, but in colon, where both arms are biopsies, it rose with inflammation, so blanket residualisation is unsafe across tissues. Two of three predictions failed and are reported in full. We release the protocol as a checklist with decision rules, with the cohorts, audit and code.

Keywords feature selection; benchmarking; reproducibility; batch effects; pre-analytical variables; transcriptomics

Key Points

- We provide a reusable audit protocol for cross-cohort feature-selection benchmarks (patient-overlap and procurement audit, cohort-composition and control-definition sensitivity analyses, correction benchmark) and apply it against predictions fixed and hash-stamped before the data were read; the article reports no new biomarker.
- A reproducibility comparison between feature-selection methods on public cohorts can change sign with the choice of cohorts, so it should be reported as an interval over all admissible cohort subsets and separately from predictive performance.
- When every cohort shares a procurement asymmetry between cases and controls, the pre-analytical confounder becomes the most reproducible signal; in kidney an immediate-early module separated biopsy from surgical tissue in 8 of 9 datasets.
- Where clinically defensible and available, a control obtained by the same procedure is the design alternative to prefer; residualising on a handling score worked in kidney but is tissue-conditional, because the module also rises with inflammation.
- In a test against predictions fixed and hash-stamped in advance, verdict instability recurred in liver, whose cohorts disagree about effects, but not in colon, whose cohorts agree, and the confounder was most prominent among the most reproducible genes where every cohort shared that asymmetry.

Introduction

Transcriptomic biomarker discovery has converged on a common recipe: assemble public case/control cohorts, select features that behave consistently across them, and report the signature with a classification performance figure. Consistency across independent cohorts is expected to filter dataset-specific noise, and a methods literature has grown around the idea, from multi-cohort meta-analysis [1] to stability selection [2]. The pitfalls of classifiers derived from high-dimensional expression data [3, 4, 5] and the impact of batch structure [6] are well catalogued.

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Two assumptions underlie the recipe: that a feature reproducible across cohorts more likely reflects disease biology, and that a benchmark comparing selection methods on a set of public cohorts measures the methods rather than the cohorts. That a method comparison depends on how it was set up is established [7, 8, 9, 10, 11, 12, 13], as is the instability of high-dimensional feature selection under resampling [14]. What is missing for cross-cohort biomarker discovery is a quantification: varying the cohort configuration is recommended practice, but reporting the resulting interval is not. A meta-analysis of 28 filter feature-selection studies found that the number of datasets, baselines and new methods alone explained 33% of the variance in reported win rates [15], and a recent framework grades which layers of differential-expression evidence reproduce across fixed cohorts [16]; neither varies which cohorts enter the comparison, and neither asks whether the most reproducible signal is the disease.

We address this as a protocol problem. Diabetic kidney disease (DKD) is a suitable first setting because its usable public cohorts are few enough to enumerate and numerous enough to resample. The protocol places four checks around a conventional benchmark, each taking the output of the last. *Step 1, audit the input.* Cohorts go in; a table of patient overlap across accessions and of the procurement route of each arm comes out, and a series that shares patients with another is excluded. *Step 2, vary the cohorts.* The benchmark is re-run on every admissible subset of the surviving cohorts, and the difference between methods is reported as a range with its sign changes rather than as one number. *Step 3, vary the control.* The case set is held fixed and the control group is replaced by one obtained the same way as the cases, and the two rankings are compared. *Step 4, benchmark the corrections.* Each correction available for whatever confounder the first three steps expose is scored on how much of it is removed and what that costs in external performance. Because findings in one disease cannot separate a property of benchmarking from a property of its cohorts, we then fixed three predictions before reading any liver or colon expression value, and report every verdict, including two failures.

Three contributions follow. First, we quantify how far a benchmark verdict depends on which cohorts are in it, and give a procedure that reports the range of the method difference together with predictive performance, so that one configuration is not generalised into a statement about the methods. Second, we show that the most reproducible signal can be a procurement confounder shared by every cohort, and compare what each available correction removes against what it costs, down to the change in the candidate list itself. Third, applying the protocol to liver and colon marks its boundary: the same score also rises with inflammation, so the correction that works in kidney must not be applied blindly elsewhere. The protocol is released as a checklist with decision rules (Table 4).

Materials and methods

The four audit steps of the Introduction map onto the subsections below in order: step 1 onto cohort assembly and audit, step 2 onto evaluation and cohort-composition sensitivity, and steps 3 and 4 onto the confounder score, corrections and control definition. Table 4 states the decision rule for each step.

Table 1 Subjects appearing in more than one retained kidney series

Series A	Series B	Shared	DKD cases	Non-biopsy controls	Other biopsy
GSE104948	GSE104954	55	5	1	49
GSE30528	GSE30529	9	5	4	0

These are the same patients profiled in two compartments, not duplicate cohorts. Other biopsy are the non-diabetic kidney-disease biopsies used as the procurement-matched comparator; folds are not independent for them either.

Cohort assembly and audit

Fifteen human DKD transcriptome series were retrieved from the Gene Expression Omnibus (GEO). Patient identifiers recovered from sample titles revealed reuse invisible at the accession level, and three series (GSE30122, GSE47183 and GSE99340) were excluded because the same patients would otherwise sit on both sides of a leave-one-dataset-out split (Supplementary File 2). Residual overlap that could not be removed is disclosed (Table 1): the two ERCB compartments GSE104948 and GSE104954 share 55 subjects, and GSE30528 and GSE30529 share 9.

Five cohorts entered the discovery rotation and two served as compartment validation (Table 2) [17, 18, 19, 20, 21]. In every cohort the cases are kidney biopsies and the controls are not; none of the five discovery contrasts uses a biopsy-matched control group. Further datasets served specific tests: GSE175759 [22], GSE162830 [23], GSE142153 [24], GSE1009 [25], GSE111154 [26], GSE20602 [27], GSE5406 (heart) [28], and the Kidney Precision Medicine Project (KPMP) single-nucleus atlas [29, 30], in which the 27 donors enrolled with chronic kidney disease (CKD) and a recorded history of diabetes are treated as an operational DKD-proxy subset rather than a registry-defined diagnosis.

Harmonisation and selection strategies

Probes were mapped to Entrez Gene identifiers, multi-gene probes dropped and duplicates collapsed by maximum mean, a rule fixed before analysis. The intersection over the four original discovery cohorts is 9,900 genes; GSE294519 was mapped into this frozen space and covers 98.5% of it. All cross-cohort operations use per-cohort gene-wise z-scores, which use no label information.

Four strategies were compared on the same gene space and folds (Supplementary Note S1 gives definitions and equations). DEG_meta combines per-cohort Welch t-statistics by weighted Stouffer's method [31]. WGCNA_hub ranks genes by intramodular connectivity in a soft-thresholded correlation network [32], as prior DKD work did [33]; it is implemented directly and departs from canonical WGCNA in the respects listed in Table 3, which Supplementary File 1 shows do not change any conclusion. RBS is cross-cohort stability selection (robust biomarker score) combining importance, within-cohort stability, cross-cohort reproducibility and perturbation robustness under a sign-concordance gate; it stands in for the stability-selection family [2] and no novelty is claimed for it. RBS_orth is RBS after the handling-score correction below.

Evaluation and cohort-composition sensitivity

Evaluation is leave-one-dataset-out (LODO): each strategy selects $K = 50$ genes on the training cohorts, a plain L_2 logistic model is

Table 2 Kidney transcriptome cohorts. Cases are biopsies in every cohort; controls are not

Cohort	Role	Compartment	Platform	Cases	Controls	Control tissue
GSE104948	discovery	glomerulus	GPL22945/GPL24120	12	26	donor, nephrectomy
GSE142025	discovery	cortex	Illumina HiSeq4000	27	9	nephrectomy
GSE294519	discovery	tubulointerstitium	GPL24676	23	13	nephrectomy
GSE30528	discovery	glomerulus	GPL571	9	13	nephrectomy
GSE96804	discovery	glomerulus	GPL17586	41	20	nephrectomy
GSE104954	compartment	tubulointerstitium	GPL22945/GPL24120	17	26	donor, nephrectomy
GSE30529	compartment	tubulointerstitium	GPL571	10	12	nephrectomy

Nephrectomy, tumour nephrectomy; donor, living kidney donor. All cohorts share the frozen 9,900-gene Entrez space. Compartment cohorts are the second compartment of donors already represented; they enter the procurement and comparator analyses but not the cohort-composition sweep, which requires independent cohorts.

Table 3 What the correlation-module comparator shares with canonical WGCNA

Element	Canonical	Used here	Status
Similarity	Pearson correlation	Pearson correlation	same
Adjacency	signed or unsigned power	unsigned power	restricted
Soft power	scale-free fit	fixed at $\beta = 6$	differs
Modules	TOM, dynamic tree cut	clustering on correlation distance	differs
Hub score	kME	connectivity kIN	differs
Eigengene	first principal component	not computed	absent

We therefore call it a correlation-module comparator and do not claim it reproduces the reference package. A 32-variant grid in Supplementary File 1 tests every departure, and the package itself was run as a second comparator (Table 5).

fitted on them, and the held-out cohort is scored. Predictive performance is the mean external AUROC over folds; reproducibility is the mean pairwise Jaccard index between the gene sets selected in different folds. The two are reported separately throughout. Every AUROC is read against a random-signature null of 300 same-size gene sets and a permuted-label null (Supplementary Note S2). Method differences carry a paired patient-level bootstrap of 200 replicates in which the whole evaluation, selection included, is repeated. The headline benchmark uses the four cohorts fixed before analysis; the composition sweep adds GSE294519 and repeats the complete benchmark on every subset of the five discovery cohorts of size 2–5 (26 subsets), with 30 inner bootstrap replicates per cohort.

Confounder score, corrections and control definition

The immediate-early gene (IEG) module (*FOS*, *FOSB*, *JUN*, *JUNB*, *EGR1*, *ATF3*, *DUSP1*, *ZFP36*, *NR4A1*, *NR4A2*, *BTG2*, *EGR2*, *EGR3*, *IER2*, *KLF2*, *KLF4*, *SOCS3*, *DUSP2*, *JUND*) gives a label-free per-sample handling score h_i, its mean z-score. Residualisation removes that axis from every gene with a slope estimated within class, because h is nearly collinear with case status and a total slope would absorb the disease difference:

$$\beta_j = \frac{\tilde{h}^\top \tilde{x}_j}{\tilde{h}^\top \tilde{h}}, \qquad x_{ij}^{\mathrm{corr}} = x_{ij} - (h_i - \bar{h})\beta_j, \tag{1}$$

where tildes denote centring inside each class. Two analyses report the cost of this correction and they are not the same run: the fold means use the headline stability configuration (inner bootstrap $B = 150$), whereas the paired uncertainty analysis resamples patients within each cohort and class, re-runs the whole leave-one-dataset-out evaluation in each of 200 replicates, and uses a reduced inner bootstrap ($B = 30$) to make that affordable. The corrections compared were none, this residualisation, RUVg with the module or with empirical controls [34], surrogate-variable adjustment protecting the design [35], and ComBat with cohort as batch [36]. Effect sizes are Hedges' g [37]. The specificity gate keeps genes with $|g| \geq 0.5$ for DKD against the other biopsy diagnoses of ERCB.

The control-definition sensitivity analysis holds the case set fixed and replaces the non-biopsy control by the remaining biopsy diagnoses of the same resource, then counts genes reaching $|g| \geq 0.5$ under each definition and compares the top of the two rankings. It does not separate clinical from pre-analytical causes; procurement is our reading of why the definitions differ.

Predictions fixed before the liver and colon data were read

Instability of reproducibility verdicts is a statistical claim and should appear in any tissue; a procurement confounder should become reproducible only where every cohort shares the same asymmetry. Two tissues separate these. In liver with metabolic dysfunction-associated steatohepatitis (MASH), procurement is asymmetric in some cohorts (needle biopsy cases, donor or surgical controls) and matched in others; in ulcerative colitis (UC), both arms are endoscopic biopsies in every cohort. Three predictions (P1–P3, stated in Results) were written after reading only series- and sample-level metadata and before any expression value was downloaded; the file's SHA-256 digest and time stamp were recorded, a second file stating post hoc analyses was hashed before those analyses ran, and a pipeline stage verifies both digests on every run (Supplementary File 7). The digests were recorded locally, so what they establish is that neither file has changed since it was written, not an externally certified date: there is no third-party registry entry. We therefore describe these as predictions stated and hash-fixed before the analysis rather than as a registered protocol.

Cases were MASH against histologically normal liver and actively inflamed UC colon against normal control colon, one sample per subject. Procurement was coded per arm from series text, sample fields and source publications before any effect was computed. Five core cohorts per tissue entered the composition sweep; remaining eligible cohorts entered only the per-cohort module test. Patient reuse across same-platform series was audited on expression values: a sample whose best partner in another series reached $r \geq 0.9999$ was called identical. The audit excluded GSE59071 (116 samples identical to GSE75214) and GSE17470 (9 identical to GSE24807). Selection used the genes shared by each tissue's core cohorts, 7,805 in liver (11 of the 19 module genes) and 15,789 in colon (all 19), and the kidney code and parameters were reused unchanged. The per-cohort effect g_{IEG}

is Hedges' g of the handling score with a 1,000-resample bootstrap interval. Post hoc, the effect was adjusted for a 13-gene inflammation score sharing no gene with the module, cross-cohort effect concordance was defined as the median pairwise Spearman correlation of gene-wise g, and colon cohorts were subsampled to kidney sizes in five replicates (Supplementary Note S10 gives the full cohort, audit and harmonisation rules).

Implementation

All analyses are implemented in Python with pinned versions and run from a single entry point whose manuscript path is checked against a dependency closure of every declared output (Supplementary Note S9). Every resampling stage takes a fixed seed.

Use of AI tools. Claude Code (Anthropic) and ChatGPT (OpenAI) assisted with code development and refactoring, proposals for control analyses, manuscript drafting and revision, and figure preparation. Numerical results were computed from the cited datasets or explicitly described simulations using the analysis code. Automated checks were used to assess numerical consistency and provenance. The authors reviewed the code, analyses and manuscript and take responsibility for the final content.

Results

Reproducibility verdicts depend on cohort composition

Absolute AUROC was uninformative: random 50-gene sets reached 0.68–0.88 depending on the held-out cohort, while permuted labels gave approximately 0.50, because the DKD programme is broadly co-expressed (Supplementary Note S2). The method comparison was no more stable. Across the ten three-cohort subsets the paired difference RBS − connectivity in cross-fold agreement spanned −0.106 to +0.170 (Figure 1). A study using GSE96804 + GSE142025 + GSE294519 would report +0.170 in favour of stability selection, and one using GSE30528 + GSE104948 + GSE142025 would report −0.106 against it; both are legitimate DKD benchmarks. RBS won on reproducibility in 9 of 10 three-cohort subsets and lost when all five cohorts were used.

External AUROC behaved differently. Its mean advantage was positive at every subset size and grew with cohort count, to +0.139 at five cohorts. The paired bootstrap put that five-cohort difference at +0.129 [−0.002, +0.263] ($p = 0.06$): the mean difference favoured stability selection, but the interval included zero. The two metrics did not even agree on the winner: excluding ties, the method first on reproducibility was first on external AUROC in 11 of 24 subsets. A benchmark reporting one of these numbers is therefore not reporting a weaker version of the other. Table 4 lists what a benchmark on public cohorts should report as a consequence.

The instability is not a property of one selector pair. Repeating the 26-subset sweep with each of six further inner selectors of stability selection, against the same connectivity comparator, reproduced it in every case (Table 5): the three-cohort reproducibility difference changed sign under all seven selectors, in 1 to 5 of the 10 subsets, and with all five cohorts the connectivity comparator won on reproducibility while stability selection won on external AUROC under every selector. Random forest was the least stable, reversing in half of the three-cohort subsets. What the seven share is the comparator and the cohorts, so the instability belongs to the cohort set rather than to any selector, within the classical families tested.

The comparator side was checked the same way. Replacing the reimplementation by the WGCNA package itself, run with its tutorial defaults on every fold (settings in the note to Table 5), reproduced the reversal: the three-cohort difference spanned −0.081 to +0.261, and with all five cohorts the package won on reproducibility (−0.133) while stability selection won on external AUROC (+0.080). The package was the weaker comparator on both axes (mean Jaccard 0.064 and AUROC 0.711 over the 26 subsets, against 0.107 and 0.783 for the reimplementation), so the reimplementation did not flatter stability selection.

The most reproducible signal tracks tissue procurement

No DKD cohort we examined reports warm or cold ischaemia time, fixation or time to stabilisation, so procurement route is read from sample annotations and its attribution to handling is an inference. It rests on three observations. Across nine kidney datasets the IEG module was lower in biopsy-procured than in nephrectomy- or donor-procured groups in eight (Figure 2; Spearman $\rho = -0.75$, $p < 0.001$); the exception, GSE162830, processed all 32 samples identically and showed no effect (−0.05). In KPMP single-nucleus pseudobulk, DKD separated from non-biopsy reference tissue ($p = 6.0 \times 10^{-76}$) but not from non-diabetic CKD, which is also biopsied ($p = 0.21$; Supplementary Note S3). And warm ischaemia induces exactly these transcripts within minutes, including in kidney [38], and drives transcriptome variation across human tissues [39]. An independent comparison of reference tissue sources within KPMP has since reported the same dependence [40]. In heart failure (GSE5406), where explanted and donor hearts are handled similarly, the module separated explant from donor only weakly ($g = -0.23$).

Because this asymmetry is shared by every discovery contrast, the confounder is maximally reproducible, and a criterion that rewards consistency ranks it first: uncorrected RBS placed 10 immediate-early genes in its top 50. The connectivity comparator placed none, but this exclusion alone does not establish robustness. The module's genes are strongly trait-correlated yet rank low by connectivity, because intramodular connectivity sums adjacencies and rewards membership of large blocks: the median block size is 19 for the immediate-early genes against 705 for the top 50 by connectivity. All 32 variants of the comparator placed no immediate-early gene in the top 50, and simulation shows the advantage reverses when a confounder forms the larger block (Supplementary Note S4). The WGCNA package itself did not always exclude the module: across the 75 leave-one-dataset-out folds of the cohort sweep, which evaluate 30 distinct training combinations, it placed at least one immediate-early gene in its top 50 in 9 folds (3 distinct training combinations), up to 18 of 50, and every such fold was one in which dynamic tree cut had produced a small trait module (61 to 199 genes, against a median of 588). The package ranks by |kME| within the chosen module, so this does not test the summed-connectivity account directly; the findings are consistent with module-dependent exclusion rather than general robustness to confounding.

Which corrections work

ComBat with cohort as batch returned the same values as no correction in this implementation and evaluation setting (Table 6, Figure 3). Cohort-level batch adjustment alone cannot disentangle procurement from disease when the two are confounded within cohorts. RUVg with empirical controls also failed, so the control

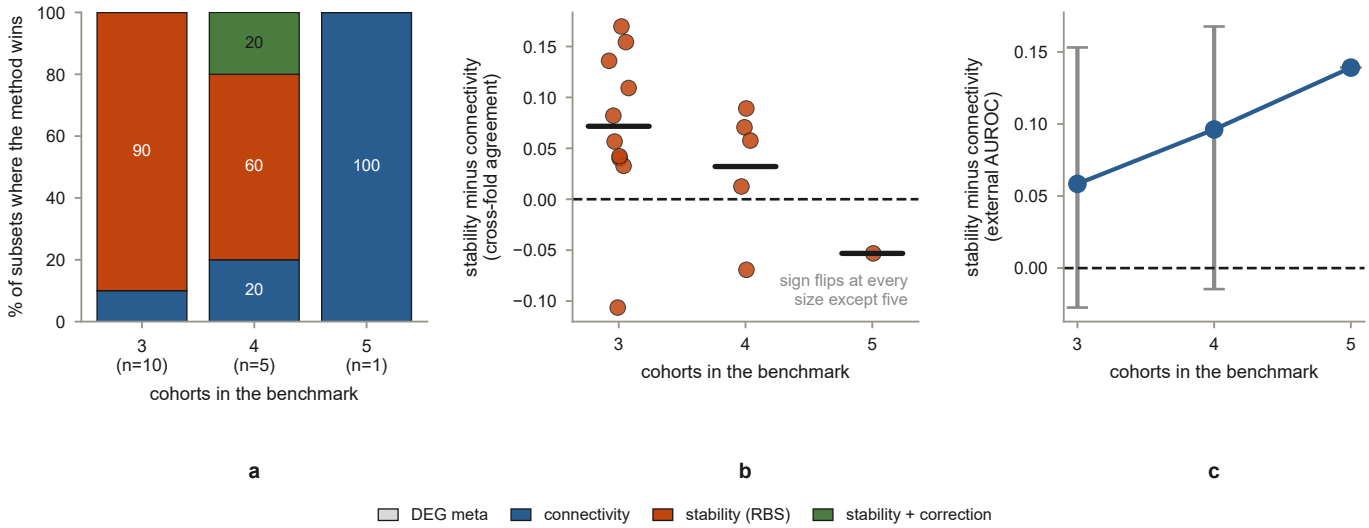


Figure 1 The reproducibility verdict depends on which cohorts are in the benchmark; the mean predictive advantage stays positive. The benchmark was repeated over all 26 subsets of the five discovery cohorts; panels show the 16 of size three or more. **a** Percentage of subsets in which each method wins on cross-fold reproducibility, ties excluded. **b** Stability selection minus the connectivity comparator in cross-fold agreement for every subset; the difference spans -0.106 to $+0.170$. **c** The same subsets scored by external AUROC, where the mean advantage is positive at every size ($+0.058$, $+0.096$, $+0.139$ for sizes three to five). Custom co-expression hub comparator; not the reference WGCNA implementation.

Alt text: Three panels. Left, stacked bars for benchmarks of three, four and five cohorts showing the percentage of cohort subsets in which each feature-selection method wins on cross-fold reproducibility; stability selection wins 90 percent of three-cohort subsets and none of the five-cohort benchmark. Middle, one point per subset of the reproducibility difference between stability selection and the connectivity comparator, ranging from minus 0.106 to plus 0.170, with points on both sides of zero at three and four cohorts. Right, the mean external AUROC difference with its range, positive at every size and rising with cohort count.

Table 4 The audit protocol as a checklist: what to report, why, and what to do when the check fails

Report	Because	What to do when the check fails	Where applied
A patient-overlap audit across accessions	reused patients make subsets and folds non-independent	exclude the duplicated series, and disclose the overlap that cannot be removed	Table 1
Procurement route of cases and of controls, per cohort	collinearity with diagnosis is a confounder no selector can remove	where no comparator shares the procurement route, hold the attribution open instead of naming a cause	Table 2, Figure 2
Pre-analytical variables to the SPREC or BRISQ standard	they turn the inference into a measurable covariate	where unavailable, say so and keep procurement as an inference	unavailable in every cohort used
The method difference as a range over all admissible cohort subsets, and whether it changes sign	a single configuration is one draw from that range	report the range, not the single verdict; with too many cohorts to enumerate, draw a pre-specified sample of configurations and say how it was drawn	Figure 1
Cross-cohort concordance of gene-wise effects	low concordance is where sign changes appeared	treat a single-configuration ranking as local to those cohorts and withhold the general claim	Figures 1, 5
Winner frequency at a fixed subset size, ties excluded, with reproducibility and predictive performance side by side	with few folds ties inflate apparent wins, and the two metrics agreed on the winner in only 11 of 24 subsets	report both axes separately; do not let one stand in for the other	Figure 1
Every AUROC with its position in a random-signature null	random 50-gene sets reached 0.68–0.88 here	if the selected set does not clear its own null, report the margin rather than the absolute figure	Supplementary Note S2
A control obtained by the same procedure wherever one exists	it did more work here than any algorithm	where an alternative control exists, compare the rankings under both definitions; where none exists, state that this sensitivity cannot be assessed and treat the candidate order as conditional	Figures 4, 5
Whether the correction score also carries disease biology	the same module rose with inflammation in colon	assess how far the score overlaps the disease signal and report the selection with and without the correction; where the two cannot be separated on the available evidence, withhold blanket residualisation and prefer the design-based control	Figure 5

The rows follow the four steps of the Methods in order: audit the input, vary the cohorts, vary the control, benchmark the corrections. The null-comparison row applies to every AUROC reported in any step. Each check was applied here; the last column says where the result appears.

Table 5 Verdict instability under each inner selector of stability selection

Inner selector	Cross-fold Jaccard, three cohorts	Neg.	External AUROC, three cohorts	Neg.	Jaccard, five	AUROC, five
ReliefF (headline)	-0.106 to $+0.170$	1	-0.027 to $+0.153$	2	-0.053	$+0.139$
Univariate F	-0.111 to $+0.150$	1	-0.098 to $+0.150$	2	-0.063	$+0.101$
LASSO	-0.127 to $+0.119$	2	-0.112 to $+0.146$	1	-0.090	$+0.078$
Elastic net	-0.163 to $+0.119$	2	-0.033 to $+0.119$	1	-0.106	$+0.091$
mRMR	-0.105 to $+0.154$	1	-0.096 to $+0.129$	2	-0.037	$+0.088$
Random forest	-0.211 to $+0.074$	5	-0.039 to $+0.127$	3	-0.144	$+0.125$
Boruta	-0.157 to $+0.120$	2	-0.106 to $+0.106$	3	-0.145	$+0.111$
Comparator replaced by the WGCNA package (ReliefF inner selector)						
WGCNA package	-0.081 to $+0.261$	1	-0.007 to $+0.283$	1	-0.133	$+0.080$

Stability selection minus the connectivity comparator, the inner selector of stability selection varied; the comparator is unchanged. Ten three-cohort subsets per row; Neg., subsets in which the difference is negative. Five, the single five-cohort benchmark. Rows above the rule use the custom co-expression hub comparator, not the reference WGCNA implementation. The last row replaces the custom comparator by the WGCNA package run with its tutorial defaults on every fold (soft power by scale-free fit, 4–20 here; unsigned TOM; dynamic tree cut; module most correlated with the trait; hubs by |kME|).

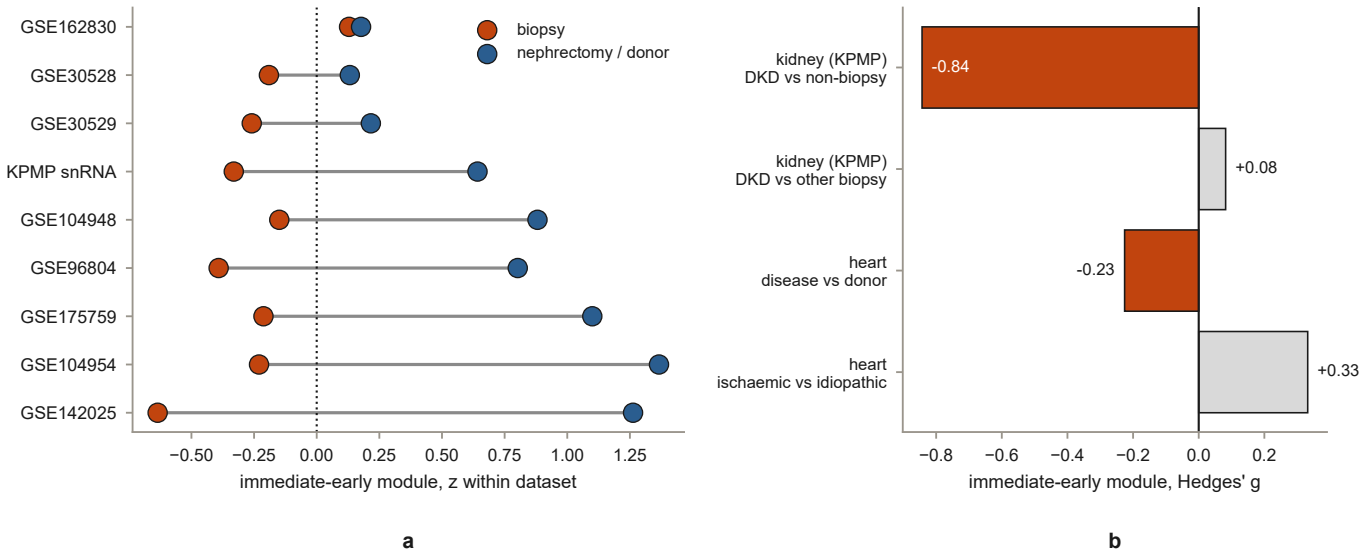


Figure 2 The most reproducible signal tracks tissue procurement. a The immediate-early module, z-scored within dataset, for every diagnosis group in nine kidney datasets. Biopsy-procured groups sit below nephrectomy- and donor-procured groups in eight of nine (Spearman $\rho = -0.75$, $p < 0.001$); GSE162830, the one identically processed design, is the exception. **b** The module as Hedges' g for four contrasts. In KPMP, DKD against non-biopsy reference tissue gives -0.84 while DKD against other biopsies gives $+0.08$; in heart failure (GSE5406), disease against donor reaches only -0.23.

Alt text: Two panels. Left, a dumbbell chart of nine kidney datasets: for each dataset the immediate-early module score of biopsy-procured groups (orange) sits below that of nephrectomy- or donor-procured groups (blue) in eight of nine; GSE162830, processed identically, shows no gap. Right, horizontal bars of Hedges' g for four contrasts: diabetic kidney disease against non-biopsy reference tissue in KPMP is minus 0.84, against other biopsies plus 0.08; heart failure against donor heart is minus 0.23, ischaemic against idiopathic plus 0.33.

set rather than the framework does the work. Residualisation and surrogate-variable adjustment reduced procurement-flagged genes from 27% to 5–6%. Table 6 reports identical fold-mean AUROCs to three decimal places. In a separate patient-stratified paired bootstrap analysis with 200 replicates (Methods), the mean uncorrected minus residualised difference was $+0.079$ (95% percentile interval -0.048 to $+0.229$). This exploratory uncertainty estimate does not establish performance preservation: a substantial loss remains compatible with the interval, whose tails are imprecisely estimated. The evaluation is not circular: the procurement flag derives from ERCB effect sizes, and correlation with the handling score separates flagged from unflagged genes at AUC $= 0.480$. Across seven inner selectors (Table 7), residualisation lowered the flagged fraction for every one, from 17.0–35.5% to 3.5–12.5%, at an AUROC cost that ranged from a gain of 0.021 to a loss of 0.114; a study adopting it should report that cost for its selector.

The control-group problem is not specific to diabetic kidney disease

The ERCB compartments contain eight biopsy diagnoses alongside the same non-biopsy controls, so the control-definition analysis can ask inside one resource whether the effect belongs to DKD or to the design. For each of seven diagnoses (thin membrane disease had too few cases), substituting the other biopsy diagnoses for the non-biopsy control retained only 14–56% of the naive median effect size (Table 8); in IgA nephropathy the 3,800 genes reaching $|g| \geq 0.5$ fell to none (Supplementary Note S6). Diabetic nephropathy ranked fourth of seven. The substitution did not merely shrink the signal; it

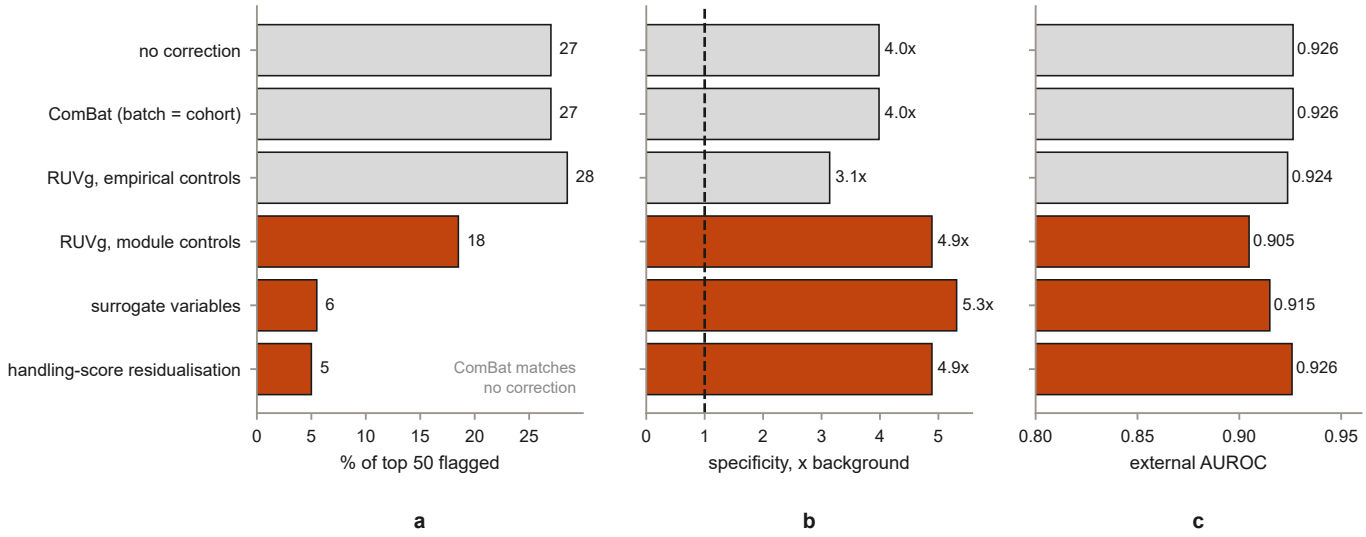


Figure 3 What each correction removes, and what it costs. a Percentage of the top 50 selected features flagged as procurement-driven. **b** Specificity pass-rate as fold-enrichment over the 8.3% background rate at which a random gene passes the gate. **c** External AUROC. Handling-score residualisation and surrogate-variable analysis reduce procurement-driven features from 27% to 5–6%; residualisation does so at a fold-mean AUROC identical to no correction to three decimal places, surrogate-variable analysis at 0.915 against 0.926. The separate paired bootstrap estimate of the residualisation cost is uncertain and is given in the text. ComBat returned the same values as no correction in this evaluation setting; the confounder is within-cohort.

Alt text: Three panels of horizontal bars for six correction strategies. Left, the percentage of the top 50 selected genes flagged as procurement-driven: 27 percent with no correction or ComBat, 6 percent with surrogate variables and 5 percent with handling-score residualisation. Middle, specificity pass rate as fold enrichment over background, all above threefold. Right, external AUROC from 0.905 to 0.926, with no correction and residualisation both at 0.926.

Table 6 Correction benchmark at $K = 50$

Correction	IEG top 50	Flagged	Spec. pass	AUROC	Cross-fold
None	8.5	27%	33%	0.926	0.284
ComBat (cohort)	8.5	27%	33%	0.926	0.284
RUVg, empirical	7.5	28%	26%	0.924	0.154
RUVg, module	0.8	18%	40%	0.905	0.215
Surrogate variables	2.0	6%	44%	0.915	0.251
Residualise on h	1.0	5%	40%	0.926	0.260

IEG top 50 and the other columns are means over the leave-one-dataset-out folds, so the uncorrected count differs from the 10 of the single headline run. Flagged, procurement-flagged share of the top 50. Spec. pass, share clearing the DKD-versus-other-CKD gate, whose background pass rate is 8.3% and permutation-based false-discovery estimate 16.7%.

Table 7 The handling-score correction under each inner selector ($K = 50$, $B = 40$)

Selector	Flagged		AUROC		Change
	before	after	before	after	in AUROC
Univariate F	24.5%	6.0%	0.921	0.807	-0.114
LASSO	33.0%	12.5%	0.956	0.879	-0.077
Elastic net	35.5%	3.5%	0.909	0.857	-0.052
mRMR	28.0%	6.5%	0.941	0.896	-0.046
ReliefF	29.5%	5.0%	0.928	0.949	$+0.021$
Random forest	17.0%	7.5%	0.895	0.815	-0.079
Boruta	22.0%	6.5%	0.918	0.901	-0.017

Flagged, procurement-flagged share of the selected features; before and after the handling-score correction. Each row is a mean over the leave-one-dataset-out folds on the four headline cohorts. The correction lowers the flagged share for every selector; what it costs in AUROC does not follow the selector's uncorrected performance, so a study adopting it should report that cost for its own selector. The ReliefF row repeats Table 6 at the lower B; the small differences are bootstrap resolution, not a change of method.

rebuilt the top of the ranking (Figure 4). The 50 largest effects under the two definitions shared 1 gene in diabetic nephropathy, while rank correlation over all 9,900 genes was 0.69, and the pipeline run with and without the handling-score correction returned 30 candidates each, sharing only 14. Which genes a study nominates is therefore set by how its control tissue was obtained as much as by the disease.

Two checks outside the discovery cohorts support the substitution. In GSE20602, which compares nephrosclerosis glomeruli against tumour nephrectomy, the candidates moved further in the diabetic contrast than in the nephrosclerosis one (median $|g_{\mathrm{DKD}}| - |g_{\mathrm{NSC}}|$ of $+1.02$ against -0.17 for the remaining genes, Mann–Whitney $p = 5.1 \times 10^{-18}$). And the comparator behaves the same way in a different molecular layer: replacing the healthy controls of the plasma metabolome ST003255 with the non-diabetic kidney disease patients of the same study dropped the random-set AUROC from 0.913 to 0.724, because the diabetic-versus-healthy and other-kidney-disease-versus-healthy effect vectors correlate at $r = 0.851$. A disease-relevant comparator separates disease-associated changes from those shared across kidney diseases; in plasma it does not, of course, establish the procurement mechanism.

Both mechanisms outside kidney: predictions fixed before the data

Verdict instability (P1). In liver the difference in cross-fold agreement between stability selection and the connectivity comparator ranged from -0.008 to $+0.174$ across the ten three-cohort subsets, crossing zero in one of the ten subsets tested and narrowly, while the external AUROC difference stayed positive in all ten (Table 9). That

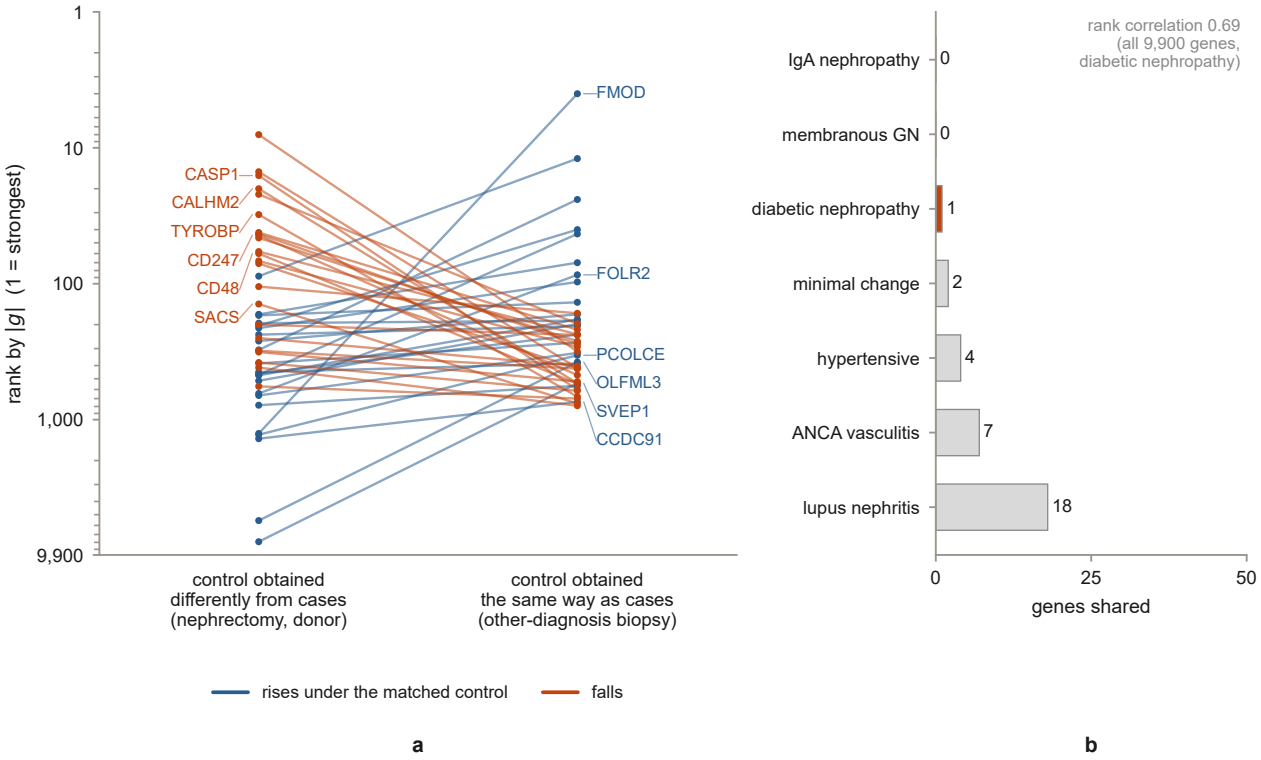


Figure 4 Changing the control-group definition rebuilds the top of the ranking. a Each line is one gene, drawn between its rank by $|g|$ when diabetic nephropathy cases are compared with non-biopsy controls (left) and with biopsies of other diagnoses (right); blue rises, orange falls. **b** Genes shared by the two top-50 lists, over all 9,900 genes, for each of the seven ERCB biopsy diagnoses. This is the same case set against two control groups, not a within-patient pairing.

Alt text: Two panels. Left, a slope chart of gene ranks in diabetic nephropathy under two control definitions, non-biopsy controls on the left and other-diagnosis biopsies on the right, on a logarithmic rank axis from 1 to 9,900; lines cross extensively, some genes falling from the top 10 to below rank 100 and others rising. Right, horizontal bars of the number of genes shared by the two top-50 lists for seven diagnoses, from 0 in IgA nephropathy to 18 in lupus nephritis, with 1 in diabetic nephropathy.

Table 8 Genes at $|g| \geq 0.5$ before and after matching the control group, by diagnosis

Diagnosis	Cases	vs non-biopsy	vs other biopsy	Effect retained
IgA nephropathy	52	3,800	0	14%
Systemic lupus erythematosus	65	4,506	166	30%
Hypertensive nephropathy	35	3,251	9	30%
Diabetic nephropathy	29	5,107	833	37%
ANCA-associated vasculitis	43	5,878	1,981	42%
Minimal change disease	27	3,351	690	54%
Membranous glomerulonephropathy	38	3,282	702	56%

All seven diagnoses share the same 52 non-biopsy controls; the matched contrast replaces them with the remaining biopsy diagnoses of the same resource. Effect retained is the ratio of median $|g|$ under the matched contrast to median $|g|$ under the naive one, an effect size rather than a count, so it does not follow the differing group sizes. Diabetic nephropathy is fourth of seven, so the design effect is not specific to it. All seven Wilcoxon tests on the paired effect vectors give $p < 10^{-300}$.

one reversal is small, so we asked whether it is an artefact of the resampling seed. Repeating the ten three-cohort comparisons under five bootstrap seeds reproduced it in the same subset (GSE89632 + GSE48452 + GSE83452) every time, at -0.017 to -0.008, while the other nine subsets stayed positive under all five seeds. The fifty runs are repeated computations on one dataset rather than fifty independent tests, so this supports robustness to computational randomness and says nothing about patient-sample uncertainty. P1 therefore holds in liver with that scope: a small reversal, repeatable, in one configuration of ten. In colon it never crossed zero ($+0.006$ to $+0.192$), and the prediction failed there. Post hoc, the tissues differed most in how far their cohorts agree about the disease: the median pairwise correlation of gene-wise effect sizes was 0.66 in colon against 0.22 in kidney and 0.15 in liver. Subsampling colon cohorts to kidney sample sizes produced only rare, small reversals: with each colon cohort subsampled to kidney case and control counts, concordance stayed between 0.60 and 0.65 in five replicates, and the difference crossed zero in two of them by at most 0.013 (Figure 5c). Sign changes were therefore associated with cohorts that disagree about the disease signal; concordance can be computed before any method is run. Three tissues cannot establish it as a necessary condition.

The module at the cohort level (P2). Where cases were biopsies and controls surgical or donor tissue, the score was lower in cases in 9 of 10 cohorts across kidney and liver, with liver GSE89632 at $g = -4.60$ [-6.11, -3.82] (Figure 5a; Supplementary Figure S5). The prediction nonetheless failed, because matched cohorts were not near zero: in colon the score was higher in inflamed cases in 9 of 10 cohorts ($+0.51$ to $+3.43$). The module also responds to active

Table 9 Predictions fixed before the liver and colon data were read, with the kidney reference computed by the same code

Prediction	Kidney, DKD	Liver, MASH	Colon, UC	Verdict
P1. The three-cohort RBS − connectivity difference in cross-fold agreement spans zero and the external AUROC difference changes sign in fewer subsets	-0.106 to $+0.170$ (1 of 10 negative) -0.027 to $+0.153$ (2 of 10)	-0.008 to $+0.174$ (1 of 10) $+0.040$ to $+0.377$ (0 of 10)	$+0.006$ to $+0.192$ (0 of 10) -0.005 to $+0.067$ (1 of 10)	held in liver, failed in colon
P2. $\lvert g_{\mathrm{IEG}}\rvert < 0.5$ in procurement-matched cohorts; negative, and larger than in every matched cohort, where cases are biopsies and controls surgical	7 asymmetric, 6 negative	3 asymmetric, all negative; 3 of 4 matched below 0.5	10 matched, 1 below 0.5, 9 positive	failed in both
P3. Immediate-early genes in the uncorrected stability top 50: kidney > liver > colon, colon at most 1; correction reduces the count substantially only in kidney	8.0 (7.2 removed)	0.6 (0.6 removed)	0.0	held

Wording abbreviated from the locally fixed prediction file (Supplementary File 7). The single negative liver subset recurs under five bootstrap seeds (-0.017 to -0.008), and the other nine stay positive in all five. Kidney values are a reference, not part of any prediction. By the same per-subset count the second clause of P1 would not hold in kidney either (2 AUROC sign changes against 1); the predictive verdict rests on the mean advantage per subset size, positive at every size in all three tissues. The kidney P3 count is a fold mean at the sweep's inner bootstrap size and so differs from the 10 of the headline run.

inflammation. Adjusting for the inflammation score shrank the colon effect by a median of 49% and left every biopsy-versus-surgical cohort negative, all ten after adjustment against nine before (median standardised coefficient -0.94 before and -1.14 after; Mann–Whitney $p = 1.6 \times 10^{-4}$ against matched cohorts). The kidney attribution survives, and the one kidney cohort with a positive raw effect turns negative once inflammation is accounted for.

Reproducibility of the module (P3). On each tissue's five core cohorts, the uncorrected stability top 50 held a mean of 8.0 immediate-early genes in kidney, 0.6 in liver and 0.0 in colon, and residualisation removed 7.2 of them in kidney (Figure 5b). Liver carries the largest procurement effects in this study, but only two of its five core cohorts are asymmetric; colon shares a direction of effect, but there the module is one inflammation-responsive set among stronger ones. Relaxing the liver expression filter raised module coverage from 11 to 15 genes and left the count at 0.6. Post hoc, control substitution within liver GSE48452 moved the module effect in the expected direction but with overlapping intervals, and is recorded as unresolved (Supplementary Note S10).

What the audit changes in a candidate list

The last step of the protocol asks what the correction does to the output it is meant to protect. Run with and without the handling-score correction, the pipeline returned 30 specificity-gated candidates each time, sharing only 14. The corrected list moved towards established biology (20 of 30 genes already proposed as DKD markers against 13 of 30), and the cell requiring a gene to be both strong in the data and absent from the literature was empty before and after the correction. Two candidates without prior DKD literature were removed by three independent tests, and protein-level corroboration in two patient-independent datasets was positive but wide and partly non-independent; Supplementary Note S7 reports it with the qualifications. We therefore report no novel biomarker, and read the value of the correction as diagnostic rather than generative: it tells a reader that a list assembled without it is substantially a different list, and that the difference does not run towards novelty.

Discussion

Cross-cohort consistency is a proxy for validity only when cohorts differ in the ways that matter. In public DKD transcriptomics they do not: every discovery contrast compares biopsies with surgical tissue, so the strongest shared signal is associated with procurement. This is a batch effect in the sense of Leek et al. [6], aggravated by being confounded with the outcome by design. More cohorts of the same design make such a confounder look more reproducible, not less. Applying the protocol to two further tissues turned this into a conditional statement: the confounder appeared wherever procurement was asymmetric, and was most prominent among the most reproducible genes where every cohort shared that asymmetry.

The instability of benchmark verdicts has a broader reach. That rankings shift with benchmark composition is not new [7, 8, 9, 10, 12]; what these data add is its magnitude in a setting where reporting a single configuration is the norm, the observation that reproducibility flips sign while the mean external AUROC advantage does not, and an association: the flips went with cohorts that disagree about the disease signal, and in the one tissue whose cohorts agreed closely, reversals stayed rare and small even after subsampling to kidney cohort sizes. Three tissues are too few to treat concordance as a necessary condition or a threshold, but it is cheap to compute before any selection is run, and a low value warns that a single-configuration verdict is unlikely to transfer.

Three practical recommendations follow (Table 4). Report reproducibility and validity as separate axes. Include a control obtained by the same procedure wherever one exists: the 310 non-diabetic CKD biopsies in ERCB did more work than any algorithm in this study. And report pre-analytical variables in the SPREC [41, 42] or BRISQ [43] format; had ischaemia time been recorded in any DKD cohort, the confounder would be a covariate rather than an inference. The handling score is also computable per sample without labels and can flag heterogeneous batches, but the cross-tissue test qualifies both that use and the correction built on it. Because the module rises with active inflammation, residualising on it in an inflamed tissue would remove disease biology with the artefact. In kidney the risk is bounded, since adjusting for inflammation made the biopsy-versus-surgical effect larger, but it is why we regard the score-based correction as tissue-conditional, and a control obtained by the same procedure

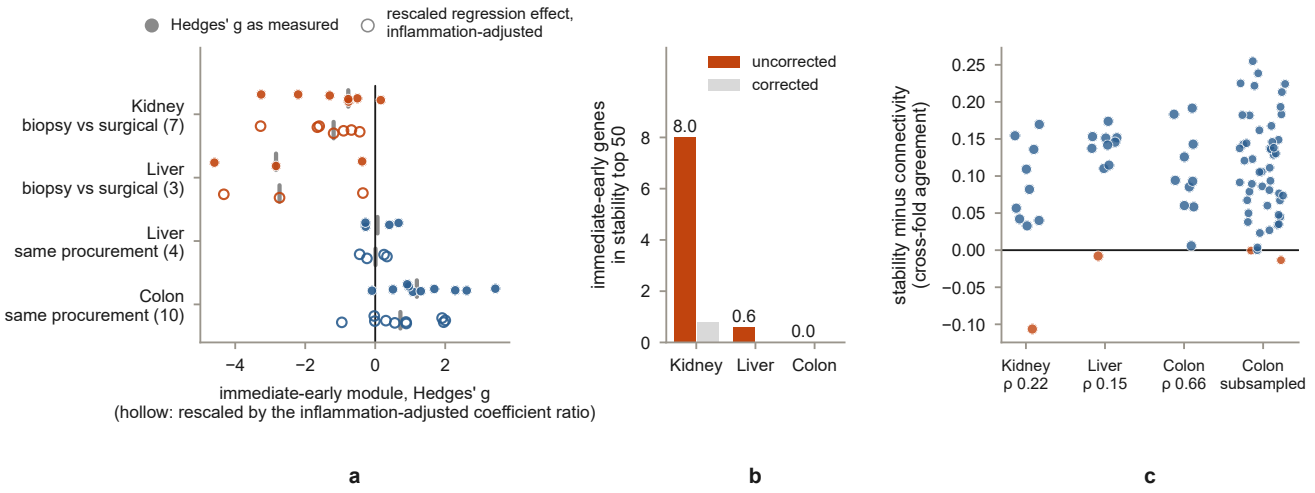


Figure 5 Outside kidney, the immediate-early module follows procurement where it is asymmetric and inflammation where it is not, and reaches the most reproducible genes mainly in kidney. a Hedges' g of the handling score, case minus control, one point per cohort, grouped by tissue and by whether cases and controls were procured differently (biopsy vs surgical) or identically (same procurement); cohort counts in brackets. Filled points are g as measured. Hollow points are not a second g: they are that g multiplied by the ratio of the adjusted to the unadjusted standardised case coefficient from the inflammation regression, which carries the adjustment onto the same axis. The two coefficients themselves, on their own scale, are in Supplementary File 7 and their medians are quoted in the text. Bars mark group medians. Per-cohort values with accession labels and bootstrap intervals are in Supplementary Figure S5. **b** Mean immediate-early genes in the stability top 50 over LODO folds on each tissue's five core cohorts, uncorrected and after residualising on the handling score. **c** Stability selection minus the connectivity comparator in cross-fold agreement for every three-cohort subset; orange points are reversed verdicts. ρ is the median pairwise Spearman correlation of gene-wise effect sizes. The rightmost column repeats colon subsampled to kidney sizes. The inflammation adjustment, concordance and subsampling are post hoc. Custom co-expression hub comparator; not the reference WGCNA implementation.

Alt text: Three panels. Left, a dot plot of the immediate-early module effect per cohort in four groups: kidney and liver cohorts with biopsy cases and surgical controls have negative values, liver cohorts with matched procurement sit near zero, and colon cohorts with matched procurement are positive; filled and hollow points show measured and inflammation-adjusted effects. Middle, bars of immediate-early genes in the stability top 50: 8.0 in kidney, 0.6 in liver, 0.0 in colon, each reduced by correction. Right, per-subset reproducibility differences for kidney, liver, colon and colon subsampled to kidney sizes, with reversed verdicts in orange: one each in kidney and liver, none in colon, two of fifty in the subsampled colon.

as the design alternative to prefer where clinically defensible and available.

Limitations

The empirical comparisons cover a limited set of established selection strategies, including a custom co-expression hub comparator, rather than an exhaustive benchmark of contemporary methods. The reversals recur under seven classical inner selectors and when the custom comparator is replaced by the WGCNA package (Table 5), so they are not an artefact of the reimplementation; the exclusion of the immediate-early module, by contrast, held for the reimplementation and its 32 variants but not for the package on every fold, and appears to depend on module detection rather than to be a property of connectivity criteria as a class. The magnitudes reported apply to the implementations and cohort configurations examined here and should not be generalised to entire method families. The audit procedure can be applied to additional selectors, but their sensitivity to cohort composition remains to be evaluated.

The confounder-matched evidence in kidney rests on ERCB and KPMP, and GSE104948 contributed both to discovery and to the specificity gate; re-deriving the candidates without it kept 17 of the 30 and left the external replication intact (Supplementary Note S8). The five discovery cohorts are few for LODO-based claims, and the composition analysis quantifies rather than removes that limitation. The simulation is deliberately simple. Outside kidney, only three liver cohorts had asymmetric procurement, one with a mixed case arm; procurement was coded from series text and publications rather than recorded handling; and the explanations offered for the two failed predictions, inflammation and effect concordance, were examined only post hoc. The central claim rests on single-modality transcriptomics: the proteomic and germline analyses corroborate or bound it, and no public resource carries a multi-layer design complete enough to form a diabetic-versus-control contrast in the same participants (Supplementary Note S8).

Conclusion

A difference in reproducibility between selection methods should be assessed together with its cohort-composition sensitivity before it is generalised: in these data the verdict changed sign across admissible subsets of the same cohorts, and the range mattered most where cohorts disagreed about effects. The most reproducible signal in such a comparison can be a pre-analytical confounder shared by design. Removing it changes which genes are reported, only 14 of 30 candidates surviving, but does not produce a defensible novel candidate: its value is diagnostic rather than generative. The checklist in Table 4 and the released code allow each check to be applied to other diseases before a candidate set is carried forward.

Supplementary data

Supplementary data are available online at *Briefings in Bioinformatics*. Supplementary Notes S1–S10 extend the methods and results; Supplementary Files 1–7 contain the comparator validation, the cohort audit, harmonisation and candidate tables, per-diagnosis results, the full candidate evidence matrix, the proteomic cohort records, and the cross-tissue test with the predictions as they were hash-fixed.

Conflicts of interest

The authors declare that they have no competing interests.

Data availability

All transcriptome cohorts are available from GEO under accessions GSE30528, GSE30529, GSE96804, GSE104948, GSE104954, GSE142025, GSE294519, GSE162830, GSE175759, GSE142153, GSE1009, GSE111154, GSE20602 and GSE5406 (kidney and heart); GSE89632, GSE48452, GSE162694, GSE126848, GSE83452, GSE66676, GSE63067 and GSE24807 (liver); and GSE75214, GSE87466, GSE179285, GSE11223, GSE47908, GSE38713, GSE53306, GSE16879, GSE48958 and GSE13367 (colon). Single-nucleus data are available from the KPMP Atlas (`https://atlas.kpmp.org/`). The protein datasets are Mendeley Data 83k89shdx5 and PRIDE PXD041884, the germline association study is GWAS Catalog GCST90179152, and the plasma metabolome is Metabolomics Workbench ST003255. The harmonised matrices, frozen gene space, phenotype tables with procurement annotation and the cohort audit database are deposited at `https://doi.org/10.5281/zenodo.23051697` [44], and all code is available at `https://github.com/kdh6126/dkd-cross-cohort-audit`.

Author contributions statement

D.K. conceived the study, designed the methodology, implemented the pipeline, curated the data, performed the analyses and validation, produced the figures and wrote the manuscript. M.B., J.P., H.L. and H.J.L. contributed to the methodology and data curation and reviewed the manuscript. S.L. acquired funding, administered the project and supervised the study. All authors reviewed and approved the final manuscript.

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